[Abstract]

FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness

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Abstract: Transformers are slow and memory-hungry on long sequences, since the time and memory complexity of self-attention are quadratic in sequence length. Approximate attention methods have attempted to address this problem by trading off model quality to reduce the compute complexity, but often do not achieve wall-clock speedup.

We argue that a missing principle is making attention algorithms IO-aware-that is, carefully accounting for reads and writes between levels of GPU memory. We propose FlashAttention, an IO-aware exact attention algorithm that uses tiling to reduce the number of memory reads/writes between GPU high bandwidth memory (HBM) and GPU on-chip SRAM.

[IO-Aware Tiling & Online Softmax Recomputation]

3 FlashAttention Algorithm: Standard Self-Attention (Vaswani et al., 2017) materializes the N x N attention matrix S = Q K^T and P = softmax(S) in slow GPU High Bandwidth Memory (HBM), requiring O(N^2) memory accesses. FlashAttention splits inputs Q, K, V into blocks, loads them from HBM into fast on-chip SRAM, and computes exact attention using online softmax scaling without ever materializing the intermediate N x N attention matrix in HBM.

During the backward pass, by storing the output O and the softmax normalization statistics (m, l) in HBM, FlashAttention recomputes the attention blocks on-chip in SRAM, reducing HBM memory complexity from O(N^2) to linear O(N) in sequence length N while producing the exact same mathematical output as standard attention.

[Wall-Clock Speedup & Benchmark Results]

4 Empirical Validation: FlashAttention trains BERT-large (Devlin et al., 2019) 15% faster than the MLPerf 1.1 training speed record, speeds up GPT-2 training by 3x over HuggingFace and 1.7x over Megatron-LM on WikiText-103, and improves long-range arena (LRA) Path-X accuracy to 61.4% by scaling context length to 16K and 64K tokens.